DATA ANALYTICS • SATELLITE OBSERVATIONS

WILDFIRE DETECTOR

Satellite-Based Wildfire Monitoring in India

2016–2025

PROJECT PROPOSAL

PROJECT TEAM

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India • Satellite hotspot observations • 10-year study

Background

Why examine historical satellite hotspot observations?

Satellite-based fire monitoring systems record thermal anomalies with geographic coordinates, acquisition dates and times, and measurement attributes. A multi-year archive makes it possible to explore when and where observations are recorded and how the recorded patterns vary across satellites and other available variables.

STUDY PERIOD

2016–2025

Ten calendar years

SOURCE FILE ROWS

854,958

Original supplied CSV

ORIGINAL FIELDS

15

Dataset variables

MISSING VALUES

0

Initial file inspection

Problem Statement

The project need

The supplied archive contains a large number of satellite observations across several years and multiple variables. Inspecting records individually makes it difficult to recognize long-term patterns, compare satellite sources, examine day and night observations, or explore geographic concentration.

PROJECT RESPONSE

- Prepare and validate the supplied dataset using a documented workflow.
- Analyze time-based, satellite-wise, category-wise and measurement-based patterns.
- Present supported findings through clear charts and an interactive Power BI dashboard.
- Distinguish observation records from unique wildfire incidents.

Objective

Main aim and measurable project objectives

Aim: Develop a data analytics and visualization project for exploring satellite-based wildfire hotspot observations in India from 2016 to 2025.

- Inspect and document the dataset and its variables; check data types, missing values, duplicates, dates and coordinates.
- Prepare a documented cleaned dataset without unjustified removal of valid observations.
- Analyze yearly, monthly and quarterly patterns; compare satellites and day/night records.
- Examine brightness, confidence, FRP, hotspot types and geographic distribution.
- Create suitable visualizations and an interactive Power BI dashboard.
- Document methods, results and limitations for a reproducible GitHub project.

Target User

Who can use the project?

STUDENTS & EVALUATORS

Review the six-stage analytics workflow, methodology, charts and dashboard outputs.

RESEARCHERS & DATA ANALYSTS

Explore historical satellite observations and compare patterns across available variables.

ENVIRONMENTAL AWARENESS GROUPS

View summarized observation patterns and geographic distributions for educational exploration.

Note: This is an exploratory dashboard, not an operational fire-alert or emergency-response system.

Proposed Technology

Tools selected for a reproducible analytics workflow

PYTHON

Data inspection, cleaning and analysis

PANDAS + NUMPY

CSV processing, transformations and calculations

POWER BI DESKTOP

Interactive dashboard, data model, filters and visuals

POWER QUERY + DAX

Data preparation and dashboard measures

MATPLOTLIB / PLOTLY

Supporting charts and interactive visualizations

GIT + GITHUB

Version control, collaboration and documentation

Questions – Variable – Chart

Matching each analytical question with suitable fields and visuals

ANALYTICAL QUESTION	VARIABLE(S)	SUITABLE CHART
Yearly change	acq_date → year	Line chart
Monthly pattern	acq_date → month	Column chart
Satellite comparison	satellite	Bar chart
Day vs night	daynight	Donut / bar
Detection categories	type	Bar chart
Brightness distribution	brightness	Histogram
Confidence distribution	confidence	Histogram
FRP variation	frp	Histogram / box plot
Geographic concentration	latitude, longitude	Map / density heatmap

Dataset Snapshot

Initial inspection of the supplied CSV and Power BI PDF

CSV RECORDS

854,958

CSV COLUMNS

15

DATE RANGE

2016–2025

EXACT DUPLICATES

0

POWER BI PDF SNAPSHOT

Total detections: 804,966 | Avg. brightness: 324.19 | Avg. confidence: 64.45 | Total FRP: 21,798,582.80

The 804,966 dashboard count matches the CSV count for type = 0; document KPI filters and definitions.

Workflow & Limitations

Six connected stages

01
Proposal
Lahari

02
Collection
Anusha

03
Cleaning
Vikasini

04
Analysis
Vikasini

05
Visualization
Anusha

06
Dashboard
Lahari

KEY LIMITATIONS

- Hotspot observations are not automatically unique wildfire incidents.
- State/district analysis requires verified boundaries and a spatial join.
- Type codes require official metadata for the exact product/version.
- The supplied Power BI PDF is a static snapshot; confirm its filters and measures.
- Verify the NASA FIRMS product URL, download settings and attribution.

Expected Outcome & Deliverable

A clear, reproducible project from source data to dashboard

The project will produce a documented dataset preparation workflow, analysis summaries, meaningful visualizations and an interactive Power BI dashboard. Users will be able to explore historical observations and compare patterns across time, satellites, day/night categories and other available variables.

GITHUB DELIVERABLE

doc/project_proposal.md

Includes background, problem statement, objective, target user, proposed technology, questions–variables–charts, dataset snapshot, workflow and limitations.

WILDFIRE DETECTOR • INDIA • 2016–2025